

# State of the Art and Technical References

Social navigation of agricultural robots in shared corridors

Thematic literature review

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**Purpose.** Present a self-contained synthesis of the literature on agricultural robotics, social navigation, proxemics, motion planning, local control, perception, and simulation instrumentation. The content is restricted to the state of knowledge and the comparative organization of the principal technical families.

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## 1 Scope of the state of the art

Social navigation for mobile robots studies how to produce safe, legible motion compatible with human expectations in shared spaces. In agriculture, this question combines with row-crop conditions, narrow corridors, vegetation-dependent perception, variable surfaces, and coexistence with workers. The literature therefore integrates motion planning, human–robot interaction, perception, control, and agricultural automation [3, 7, 8].

Contemporary surveys organize the field around three relationships: the social requirements expressed by a trajectory, the algorithmic mechanisms that generate motion, and the criteria used to interpret observed behavior [12, 9]. The agricultural domain adds the need to preserve work continuity inside strongly structured geometry and to coordinate navigation with sensors operating under illumination changes, occlusion, and dense vegetation [23, 28, 29].

**Table 1:** Themes of the literature review.

| Theme                          | Question in the literature                                     | Representative concepts                                               |
|--------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------|
| Agricultural environment       | How is navigation performed in structured crops and corridors? | Rows, orchards, plant perception, localization, and shared passage.   |
| Human interaction              | How is interpersonal space represented?                        | Proxemics, orientation, legibility, conventions, and relative motion. |
| Planning                       | How are global routes and local reactions connected?           | Navigation graphs, velocity sampling, and local optimization.         |
| Social avoidance               | How is response to people modeled?                             | Social DWA, ORCA/RVO, social force, proxemic fields, and CBF.         |
| Perception and instrumentation | How is homogeneous evidence obtained?                          | LiDAR, vision, sensor fusion, simulation, and telemetry.              |

## 2 Agricultural robotics and navigation in structured crops

### 2.1 From field automation to shared mobility

Agricultural mobile robotics has traditionally focused on row following, inspection, spraying, transport, and assisted harvesting. These tasks require the vehicle to localize itself relative to the crop and maintain a stable trajectory without damaging plants or infrastructure. Adding people to the operating space transforms geometric navigation into an interaction problem: the robot must recognize that a human obstacle has an orientation, a possible intention, and spatial needs distinct from those of a static object [1, 3].

Vineyard and orchard corridors provide a structure that favors longitudinal planning but reduces lateral alternatives. Navigation in this context combines a principal route with local adjustments for encounters, overtaking, and crossings. Recent orchard systems integrate LiDAR, vision, SLAM, and hybrid planners to preserve the navigation reference within dense vegetation and partially observable geometry [28, 29].

### 2.2 Perception and sensor fusion

LiDAR provides direct geometric measurements and is widely used to detect plant structure, estimate distance, and construct spatial representations. Vision complements geometry with semantic information, while sensor fusion seeks to preserve a coherent estimate when one modality loses quality [23, 24]. In social navigation, this integration also supports person detection, motion

estimation, and interaction-aware layers.

Recent literature on ROS architectures and autonomous navigation shows a trend toward modular systems: perception, localization, mapping, global planning, and local control exchange states through defined interfaces [27]. This separation makes it possible to compare planning families and adapt perception to the crop without redesigning the complete system.

**Table 2:** Perception functions emphasized in agricultural mobile robotics.

| Function            | Information obtained                          | Navigation use                                              |
|---------------------|-----------------------------------------------|-------------------------------------------------------------|
| Geometric detection | Distances, contours, and free space           | Construction of traversable space and immediate avoidance.  |
| Row extraction      | Crop axes and corridor orientation            | Longitudinal tracking and lateral correction.               |
| Localization        | Robot pose relative to the map                | Global-route continuity and return to points of interest.   |
| Human detection     | Estimated position, orientation, and velocity | Activation of socially aware behavior.                      |
| Sensor fusion       | Combined state from several modalities        | Greater perceptual continuity across environmental changes. |

### 3 Social navigation and human–robot interaction

#### 3.1 General principles

Human-aware navigation expands collision avoidance through social conventions. A trajectory may be physically safe yet difficult to interpret if it passes too close, changes direction abruptly, or enters a space the person considers reserved. Surveys emphasize safety, comfort, naturalness, legibility, predictability, and efficiency as complementary dimensions [3, 12, 7].

Legibility connects the form and meaning of motion. Gradual deceleration can communicate yielding; an early deviation can indicate an intention to pass; a stable trajectory makes the robot’s continuation easier to anticipate. Research on socially expressive trajectories and interaction-type-aware frameworks develops this relationship between control and nonverbal communication [17, 18].

#### 3.2 Contemporary taxonomies

The literature groups methods according to the origin of their knowledge and the way commands are produced. Model-based approaches encode explicit rules, costs, or restrictions. Interaction-based approaches represent people through forces, admissible velocities, or spatial fields. Learning-based approaches obtain policies from demonstrations or rewards. Hybrid systems combine these sources to retain interpretability and adaptation [8, 26].

**Table 3:** Conceptual organization of social navigation.

| Perspective            | Central representation                    | Decision form                                                |
|------------------------|-------------------------------------------|--------------------------------------------------------------|
| Social rules           | Thresholds, priorities, and conventions   | Discrete selection of behaviors.                             |
| Optimization           | Geometric, dynamic, and social costs      | Choice of the trajectory or command with the best objective. |
| Multiagent interaction | Relative velocities and mutual prediction | Coordination in velocity space.                              |
| Social fields          | Potentials or forces around people        | Continuous modulation of direction and speed.                |
| Formal safety          | Mathematically defined safe sets          | Command filtering through restrictions.                      |
| Learning               | Policy obtained from data or rewards      | Action inferred from environmental observations.             |

## 4 Proxemics and representation of human space

### 4.1 Interpersonal space

Proxemics describes the social organization of space around people. The classical formulation distinguishes intimate, personal, social, and public regions [2]. In robotics, these regions are interpreted as references for modeling approach cost rather than universal borders. Their meaning changes with context, task, culture, body orientation, relative velocity, and available space [3, 13].

Isotropic models assign the same reach in every direction. Anisotropic models distinguish front, side, and rear regions, reflecting that a person’s orientation changes the perception of an approach. This representation lets a planner favor clear lateral passage and avoid unnecessarily close frontal crossings [13].

**Table 4:** Proxemic levels as a conceptual framework.

| Region   | Reference separation | Interpretation in social navigation                          |
|----------|----------------------|--------------------------------------------------------------|
| Intimate | Up to 0.45 m         | Close-contact space requiring spatial priority.              |
| Personal | 0.45–1.20 m          | Sensitive area during encounters, crossings, and overtaking. |
| Social   | 1.20–3.60 m          | Common separation for functional interaction.                |
| Public   | More than 3.60 m     | Distant interaction with low immediate spatial pressure.     |

### 4.2 Orientation, motion, and groups

Body orientation introduces asymmetry: a frontal approach is detected and negotiated differently from a rear passage. Relative motion adds a temporal dimension because the same separation represents different situations when agents approach, move away, or travel in parallel. Social models therefore combine position, direction, and velocity to construct costs that evolve during interaction [13, 17].

In settings with several people, individual fields can overlap. The literature studies aggregation, saturation, and group representation so that the robot does not cross social formations or oscillate between opposing influences [8]. In agriculture, corridor geometry makes coordination between crop structure and the distribution of people especially relevant.

## 5 Global planning and local control

### 5.1 Hierarchical architecture

A mobile navigation architecture generally separates the global route from local decisions. The global planner connects origin and destination on a map; the local controller transforms that guide into commands compatible with obstacles, dynamics, and moving agents. This division appears repeatedly in ROS systems, hybrid planners, and agricultural platforms [21, 27, 29].

The global route supplies topological continuity and avoids unnecessary detours. The local layer responds to events that are not always represented on the map, such as a person entering a corridor or changing speed. Social navigation adds human costs or restrictions to this local layer while preserving the route objective and modifying immediate motion.

### 5.2 Sampling- and optimization-based local planners

The Dynamic Window Approach (DWA) evaluates velocity commands that the robot can reach dynamically and scores them according to progress, separation, and speed [6]. Its social extension introduces terms associated with human space so that command selection considers both the objective and the interaction.

Timed Elastic Band (TEB) represents a sequence of poses and times that is deformed through optimization to satisfy geometry, dynamics, and obstacles [10]. Predictive-control approaches formulate an optimization problem over a horizon and combine tracking, smoothness, and avoidance in a common objective [11]. The literature connects these families through local prediction and weighted objectives.

**Table 5:** Local-planning families described in the literature.

| Family       | Main mechanism                               | Conceptual contribution                               |
|--------------|----------------------------------------------|-------------------------------------------------------|
| DWA          | Sampling of reachable velocities             | Dynamic reaction with direct command evaluation.      |
| TEB          | Space–time optimization of a band            | Integration of geometry, timing, and smoothness.      |
| VO/ORCA      | Exclusion of velocities leading to collision | Reciprocal coordination among moving agents.          |
| Social force | Sum of goal and interaction forces           | Continuous and interpretable collective motion.       |
| CBF          | Restrictions that preserve a safe set        | Formal separation between nominal control and safety. |
| Learning     | Policy inferred from observations            | Adaptation to complex interaction patterns.           |

## 6 Dominant families of social avoidance

### 6.1 Velocity obstacles and reciprocity

Velocity obstacles represent the set of commands that would lead to a future collision under a motion model [14]. ORCA/RVO constructs admissible regions in velocity space and distributes the avoidance maneuver among agents [5]. This formulation is influential in multiagent navigation because it converts collision prediction into a geometric velocity-selection problem.

In human–robot interaction, reciprocity is interpreted as an expectation of coordination. The robot estimates that the other agent will also adjust its motion. Social-navigation surveys place this family beside joint-trajectory and motion-prediction models, where the quality of coordination depends on the representation of human response [12, 8].

## 6.2 Social force model and proxemic fields

The social force model describes motion through a force toward the destination and interaction terms associated with people and obstacles [4]. Its continuous formulation makes it possible to interpret how each influence changes the trajectory. Socially aware robotic variants adapt intensity, reach, and direction to produce gradual separation.

Proxemic fields share the idea of continuous spatial influence but emphasize the structure of interpersonal space. Anisotropy adjusts the field using body orientation and distinguishes frontal, lateral, and rear approaches [13]. This family connects proxemic theory with costs that can be incorporated into local planners.

## 6.3 Control barrier functions

Control Barrier Functions (CBF) define a safe set of states and filter control to preserve that set during system evolution [15, 16]. In navigation, a barrier can represent separation from people or obstacles and act on a nominal command generated by another policy. This modular organization has encouraged combinations with dynamic sampling and predictive control [11].

## 6.4 Hybrid and learning-based approaches

Hybrid approaches combine a geometric route with a reactive policy, or a learned policy with an explicit safety layer. Recent deep-reinforcement-learning literature addresses perception, prediction, and navigation in dynamic environments [20, 26]. In parallel, hybrid RRT–DWA and Hybrid A\*–DWA systems demonstrate the continuity of hierarchical design within autonomous architectures [21, 29].

# 7 Simulation, instrumentation, and evaluation in the literature

## 7.1 Simulation for agricultural interaction

Simulation can represent crop geometry, human agents, sensors, and controllers within repeatable scenarios. In agricultural robotics and HRI it is used to observe trajectories, study nonverbal communication, and compare behaviors before physical integration [1]. Three-dimensional engines support corridor geometry, occlusion, and human motion, while analysis tools process generated telemetry.

The state of the art tends toward instrumented architectures that separate execution and analysis. This separation supports homogeneous records of pose, velocity, distance, and interaction and enables reconstruction of temporal behavior. In modular systems, telemetry also connects simulation, robotic middleware, and external processing tools.

## 7.2 Evaluation dimensions

Social-navigation evaluation guidelines recommend reporting several dimensions instead of reducing behavior to one indicator [9, 7]. Safety describes separation and absence of contact; comfort addresses spatial intrusion and motion; efficiency considers time and length; naturalness observes fluency; and legibility examines how easily a person can anticipate the robot’s intention.

**Table 6:** Recurring evaluation dimensions in social navigation.

| Dimension              | Observation target                     | Examples of evidence                                 |
|------------------------|----------------------------------------|------------------------------------------------------|
| Safety                 | Physical separation during interaction | Distance, proximity, and contact events.             |
| Social comfort         | Respect for interpersonal space        | Zone occupancy and passing orientation.              |
| Efficiency             | Progress toward the task               | Duration, length, and movement continuity.           |
| Naturalness            | Kinematic fluency                      | Speed changes, curvature, and stops.                 |
| Legibility             | Clarity of motion intention            | Anticipation, yielding, and directional consistency. |
| Operational robustness | Preservation of expected behavior      | Behavior across scenarios and configurations.        |

## 8 Synthesis of the state of the art

The literature shows convergence toward hierarchical, socially informed architectures. Global planning preserves connectivity inside the crop; the local layer adapts commands in response to obstacles and people; perception supplies geometry and human state; and instrumentation organizes temporal evidence. This structure allows different control families to be incorporated without replacing the complete environmental representation.

**Table 7:** Comparative synthesis of the principal research streams.

| Research stream         | Contribution to the state of the art                              | Emphasized application context                             |
|-------------------------|-------------------------------------------------------------------|------------------------------------------------------------|
| Agricultural navigation | Integrates localization, crop perception, and corridor following. | Vineyards, orchards, row crops, and autonomous inspection. |
| Robotic proxemics       | Translates interpersonal space into navigation regions or costs.  | Encounters, crossings, following, and shared spaces.       |
| Social DWA              | Adds human costs to dynamic command sampling.                     | Local reaction with kinematic restrictions.                |
| ORCA/RVO                | Models reciprocal coordination through admissible velocities.     | Multiagent interaction and dynamic flows.                  |
| Social force            | Represents goals and interaction through continuous influences.   | Collective motion and interpretable control.               |
| Anisotropic fields      | Adjusts spatial influence using human orientation.                | Frontal approaches and lateral passing.                    |



| Research stream         | Contribution to the state of the art                                | Emphasized context                                      | application |
|-------------------------|---------------------------------------------------------------------|---------------------------------------------------------|-------------|
| CBF                     | Filters commands to preserve formal safety conditions.              | Safety layers over nominal controllers.                 |             |
| Learning and hybrids    | Combines data-driven adaptation with geometric or safety structure. | Dynamic environments with complex interaction patterns. |             |
| Instrumented simulation | Produces repeatable scenarios and comparable telemetry.             | Development, integration, and predeployment evaluation. |             |

Overall, the state of the art places agricultural social navigation at the intersection of four fields: mobile autonomy, human–robot interaction, environmental perception, and precision robotics. Foundational works contribute geometric and dynamic models; recent surveys contribute taxonomies and evaluation criteria; and agricultural applications demonstrate the integration of sensors, middleware, and hybrid planners. The common trend is to produce motion that preserves the task while expressing a spatial relationship understandable to people.

## 9 Conclusions

Accumulated knowledge describes agricultural social navigation as a hierarchical planning problem supported by perception and explicit representation of human space. Proxemics provides a framework for interpreting interpersonal separation; DWA, TEB, ORCA/RVO, social force, anisotropic fields, and CBF provide different mechanisms for generating or filtering motion; and hybrid architectures connect these mechanisms with global routes and agricultural sensors.

The literature review also shows that social evaluation is organized into several dimensions: safety, comfort, efficiency, naturalness, legibility, and operational robustness. This multidimensional view makes it possible to describe robot behavior technically without reducing it to a single property and provides a coherent conceptual basis for studying navigation in vineyards and other shared agricultural corridors.

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